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Counterfactual discriminative micro-expression recognition

Yong Li^{1,2*} , Menglin Liu³, Lingjie Lao³, Yuanzhi Wang³  and Zhen Cui³ 

Abstract

Micro-expressions are spontaneous, rapid and subtle facial movements that can hardly be suppressed or fabricated. Micro-expression recognition (MER) is one of the most challenging topics in affective computing. It aims to recognize subtle facial movements which are quite difficult for humans to perceive in a fleeting period. Recently, many deep learning-based MER methods have been developed. However, how to effectively capture subtle temporal variations for robust MER still perplexes us. We propose a counterfactual discriminative micro-expression recognition (CoDER) method to effectively learn the slight temporal variations for video-based MER. To explicitly capture the causality from temporal dynamics hidden in the micro-expression (ME) sequence, we propose ME counterfactual reasoning by comparing the effects of the facts w.r.t. original ME sequences and the counterfactuals w.r.t. counterfactually-revised ME sequences, and then perform causality-aware prediction to encourage the model to learn those latent ME temporal cues. Extensive experiments on four widely-used ME databases demonstrate the effectiveness of CoDER, which results in comparable and superior MER performance compared with that of the state-of-the-art methods. The visualization results show that CoDER successfully perceives the meaningful temporal variations in sequential faces.

Keywords: Affective computing, Micro-expression recognition, Temporal variation, Counterfactual reasoning, Causal graph

1 Introduction

Micro-expressions (MEs) are involuntary and rapid facial movements that reveal emotions of people in a hidden manner [1, 2]. In view of the short duration (1/25 to 1/3 second) and crypticity, MEs are quite difficult to be identified and recognized [3–12]. In contrast, macro-expressions can be relatively well accessible, but conventional (macro-)expression methods cannot be well transferred for ME because of the specialties of ME, i.e., short duration, involuntariness, and low intensity. Currently, a number of methodologies have been proposed for micro-expression recognition (MER), and most works have con-

centrated on data augmentation [4, 13], transfer learning [3], and ME dataset collection [5, 14–16].

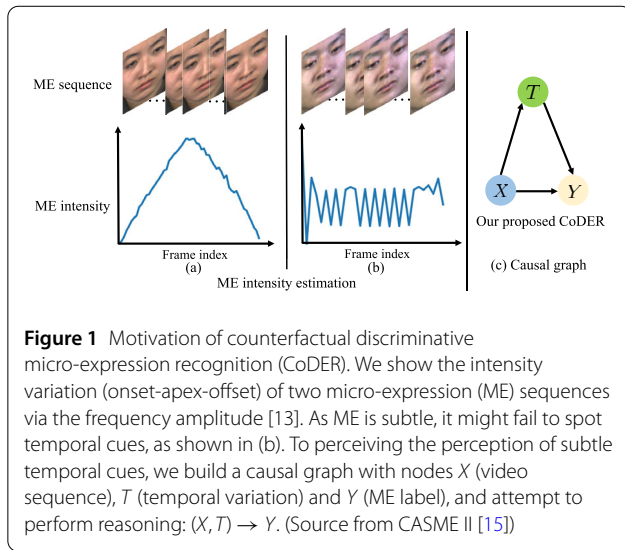
Currently, MER is challenging because of the scarcity of ME samples and the subtlety of facial deformations [5]. In the actual training phase, the network model easily overfits those undesired non-expression nuisances (e.g., head pose and facial identity) as the expression cues are too subtle. To address this problem, several methods [13, 17] have attempted to grasp the apex frame of ME, i.e., the most crucial frame in one ME sequence. It has been observed in Ref. [17] that employing the apex frame can achieve considerable performance, but there is still a gap compared with the methods employing the information from the whole ME sequence. Notably, Ekman and Friesen [18] reported that MEs were typically embedded in facial movements and more ME cues existed in sequential facial frames [19–22]. As shown in Fig. 1 (a), the intensity of ME gradually reaches the peak and then decreases, so the hid-

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den and subtle emotion can be perceived and recognized if we can explicitly capture its temporal dynamic variations. However, the dynamic cues of ME are quite subtle under the heavy influence of external face appearance, pose motion, and the other factors. It might not always be guaranteed to perceive the temporal variations of an input ME sequence manually, as illustrated in Fig. 1(b). However, temporal cues are especially important for MER even though they are rather imperceptible [5, 18, 21, 22]. Existing methods [23–26] attempt to encode temporal ME variations in two ways: 1) pre-process (e.g., optical flow [23] and dynamic image [24]) and 2) recurrent neural network (RNN) [25] and 3D convolution [26]. Specifically, we note that estimated optical flow and dynamic images are highly sensitive to head poses, thus failing to effectively capture the subtle motion information related to MEs. Furthermore, the use of 3D convolution or RNN alone is inadequate for accurately perceiving the temporal dynamics of ME. This inadequacy arises from the intrinsic subtlety of ME. Thus, we conjecture that explicitly spotting and perceiving temporal information would be more effective in capturing the nuances of these expressions. As shown in Fig. 1(c), to explicitly perceive the subtle temporal cues in ME, we attempt to make ME reasoning with $(X, T) \rightarrow Y$, where X is the input facial sequence, T denotes the temporal relation and Y is the ME label. The final prediction in counterfactual discriminative micro-expression recognition (CoDER) depends not only on the input video sequence but also on the hidden temporal relation of ME.

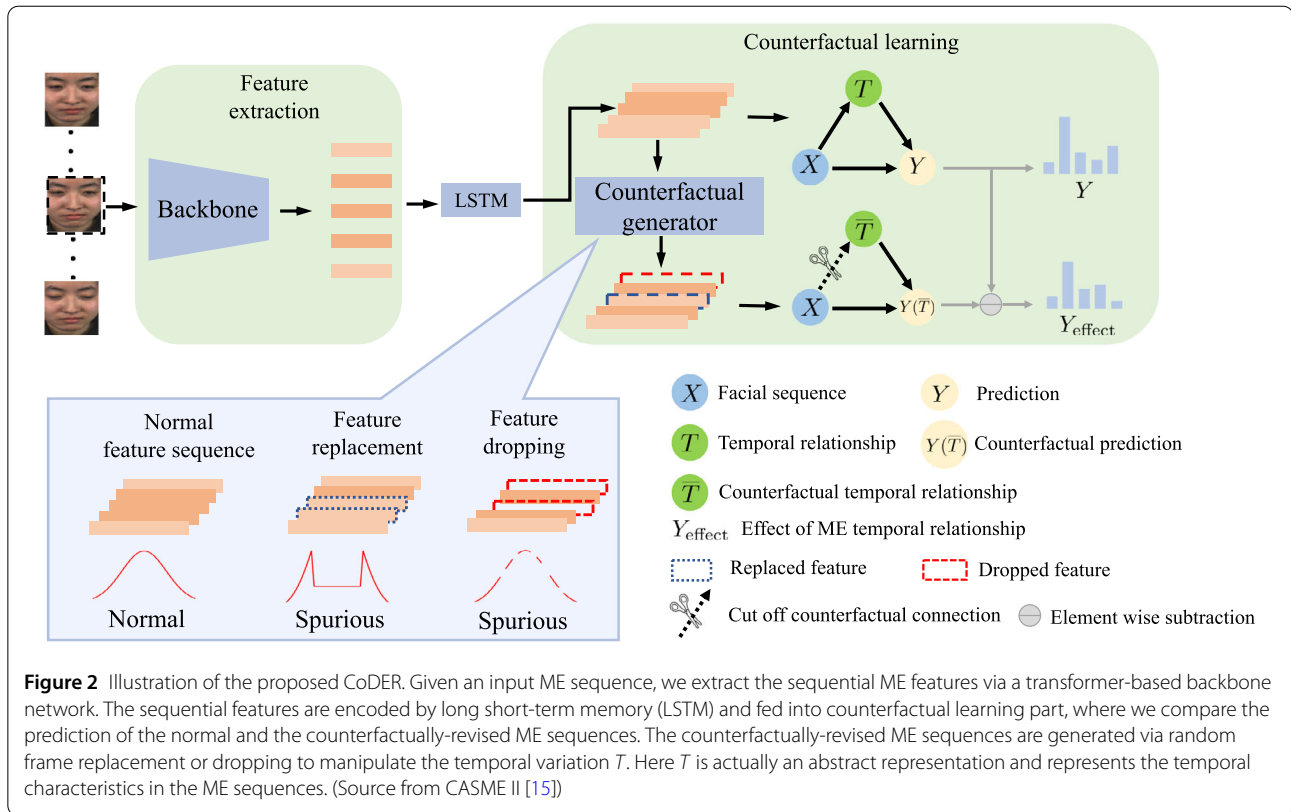
We propose a CoDER method to capture the temporal variations of ME in the video sequence. To exploit the causality hidden in the ME sequence, we propose ME counterfactual causality to analyze the effects of the hidden temporal cues explicitly. The basic idea is to reveal the influence of ME temporal relation by comparing the effects

of the original and counterfactually-revised ME sequences on the final prediction. To construct the counterfactual sequences, a counterfactual generator is built to generate numerous discontinuous ME sequences by random frame replacement or dropping within an ME sequence. We subtract the counterfactual prediction from the original prediction as the causality-aware prediction to encourage the model to learn the desired temporal cues. In this manner, the model learns to capture the subtle temporal variation of ME by spotting the difference (normal vs. spurious temporal variation) between the normal and counterfactually-revised ME sequences.

In summary, our main contributions are three-fold: 1) To the best of our knowledge, this is the first attempt to introduce counterfactual learning for MER, which takes full advantage of counterfactual sequences to encourage the model to capture those subtle temporal ME variations; 2) We propose a systematic MER framework that consists of ME representation and counterfactual reasoning, which is beneficial for small-scale dataset learning; 3) We report the state-of-the-art and comparable results on four widely-used ME databases, i.e., MMEW [5], SAMM [16], CASME II [15], and SMIC [14].

2 Related work

Micro-expression recognition. In recent years, a number of works have been proposed for MER. Many hand-crafted features, such as, local binary pattern histograms from three orthogonal planes (LBP-TOP) [27], histogram of oriented optical flow (HOOF) [28], main directional mean optical-flow (MDMO) [29], and sparse MDMO [30], are used to represent the input facial sequence. However, these shallow hand-crafted features cannot be used to perceive the subtle facial moments precisely. With the development of deep learning in computer vision, ME recognition has improved considerably. Kim et al. [22] utilized the convolutional neural network (CNN) as a feature extractor and adopted long short-term memory (LSTM) to encode the temporal characteristics of ME. Khor et al. [21] stacked the input facial sequence at the channel and feature level to obtain the robust ME representation, and then used LSTM to perceive the temporal characteristics for MER. Recently, several representative ME datasets [5, 14–16] have been publicly released to facilitate robust ME recognition. As ME video sequences are quite limited, many works propose the augmentation of training data and the use of transfer learning to increase MER performance. For example, Lei et al. [13] proposed the magnification network to generate frames of different intensities and used a graph network to encode the relationships of facial regions for precise ME classification. Xie et al. [4] utilized the generative adversarial network (GAN) to generate more training samples for MER. Since the commonality of macro-expression and micro-expression, Bin et



al. [3] introduced transfer learning, i.e., using the macro-expression dataset [31, 32] to obtain a deep model, and then transferred the knowledge in the pre-trained model to the ME network. Moreover, Sun et al. [33] proposed a teacher-student correlative framework that distilled the macro-expression knowledge to guide the ME network. Different from the existing methods, our proposed CoDER focuses on perceiving the subtle temporal variation and characteristics that naturally exist in the ME sequence by comparing the effects of facts and counterfactuals for the final MER.

Causal reasoning in computer vision. Unlike traditional inference, causal reasoning [34] focuses on analyzing the association between variables and the final results. Recently, causal reasoning has gained much attention and has been successfully used in several areas including reinforcement learning [35] and fairness [36]. Causal reasoning has been applied in many computer vision tasks, such as semantic segmentation [37], fine-grained classification [38] and human trajectory prediction [39]. Specifically, Zhang et al. [37] investigated the cause of ambiguous boundaries in weakly-supervised semantic segmentation and proposed a structural causal model to analyze the causalities among images, contexts and labels. Rao et al. [38] analyzed the bias of both intrinsic attributes and external environments on the fine-grained dataset, and proposed a counterfactual causality attention tool to measure

attention quality and guided the network to generate high-quality attention. Chen et al. [39] adopted the counterfactual analysis method for human trajectory prediction and alleviated the negative effects caused by environmental bias. Causal reasoning is an elegant and unique method to promote the status of the key variable and eliminate training bias, whereas traditional methods treat the network as a black box. We utilize causal reasoning to analyze the effect of the temporal relation in the MER task and encourage the network to highlight the temporal variations in the ME sequence.

3 ME counterfactual learning

In this section, we first provide an overview of our proposed CoDER, and then introduce crucial modules, including the ME feature extractor, causal graph construction, and counterfactual reasoning.

3.1 Overview

In Fig. 2, we illustrate the framework of our proposed CoDER, which consists of two main parts: ME feature extraction and ME counterfactual learning. We adopt transformer-based architecture [40] to extract the frame-wise feature. After ME feature extraction, we employ LSTM [41] to encode the high-level temporal variations of the sequential ME. The ME feature sequence is subsequently fed into the counterfactual learning part to en-

courage the model to perceive the temporal variations via comparison of the factual and counterfactual prediction. The corresponding details can be found in the construction of the ME causal graph in Sect. 3.3 and the ME counterfactual reasoning in Sect. 3.4.

3.2 Feature extraction

Inspired by the success of transformer in computer vision [40, 42, 43], we adopt the transformer-based backbone to extract the ME features of each frame, leveraging its elegant self-attention mechanism designed for learning contextualized representations. In detail, the original frame image is fed into a CNN backbone to extract the spatial ME feature $x_i \in \mathbb{R}^{C \times H \times W}$ for the i -th frame in an ME sequence, where C , H , and W denote the channel, height, and width. Then x_i is reshaped as $x_i \in \mathbb{R}^{C \times F}$, where $F = H \times W$ denotes the number of input tokens for the transformer module, and each token has a C dimensional feature. The transformer module contains four layers, and each layer has two main components with normalization layers between multi-head attention and feed forward network [40]. In terms of transformer-based feature extraction manner, the model is capable of attending to facial features from different information subspaces at different spatial positions.

3.3 ME causal graph

We formulate the MER task as the causal graph $\mathcal{G} = \{\mathcal{V}, \mathcal{E}\}$, where $\mathcal{V}$ and $\mathcal{E}$ are the node set and edge set. As shown in Fig. 1 (c), the node set $\mathcal{V}$ denotes the causal variables, including the input video sequence (feature) X , the temporal relationship T and the final prediction Y ; the causal link indicates the hidden causal relationships and describes how these variables interact with each other. The link $X \rightarrow Y$ represents that the input sequence directly determines the final prediction which is unwise and might not make full use of the temporal relationships, whereas $X \rightarrow T$ represents the temporal information hidden in the input video sequence. To identify the temporal variations in ME, the solution in our proposed CoDER is to perform $(X, T) \rightarrow Y$, which indicates that both the video sequence and the temporal information jointly make the final decision. Hence, in this link, we can call node X the parent of nodes T and Y , and node Y the child of nodes X and T . In ME causal inference, T plays an important role in the final prediction, because one can recognize the category of subtle ME more accurately only when the changes in facial muscles in the video are observed continuously. As a result, we can convert it as a confounder to encourage the learning of the normal temporal relationship, which will be discussed as counterfactual learning in the following section.

3.4 ME counterfactual reasoning

To take advantage of the natural causality between the continuous temporal relationships for MER, we propose

a counterfactual learning paradigm to compare the effects of the original sequences and counterfactually-revised ME sequences to encourage the ME model to realize the importance of temporal correlations in the dynamic variation process of ME.

According to the causal graph illustrated in Fig. 1(c), we analyze the effects of different variables in each link. Specifically, we can modify the attributes of certain nodes and observe the final influence, which is formally termed intervention. An idiomatic symbol $do(\cdot)$ is taken to denote the intervention. When the intervention is imposed on a certain variable, all incoming links of this variable are wiped out and a prepared pseudo value can be assigned to the variable. Moreover, the other variables usually maintain their original values. The effect of this intervened variable is found when the intervention causes a change in the final decision. For example, we can perform an intervention $do(T = \bar{T})$ to destroy the original ME temporal relationship and artificially create a new temporal relationship $\bar{T}$ as a confounder.

Here we investigate the effect of the temporal relationship T , which plays a crucial role in distinguishing subtle expressions. The external intervention is generated by destroying the normal temporal relationships on the feature sequences, thus several counterfactual sequences are obtained. Then, the final counterfactual prediction is formally estimated by a classifier layer

$$Y(\bar{T}) = f(do(T = \bar{T}), X), \quad (1)$$

where f is a prediction function such as a fully connected linear regression. In practice, we design two counterfactual generation manners, random frame replacement (R-Counter) and random frame dropping (D-Counter), as displayed at the bottom of Fig. 2. From the perspective of human visual perception, the MEs are difficult to categorize when some frames are dropped or replaced, because the essential subtle continuity of ME has been destroyed. This means that the two types of counterfactuals are obviously different from the original sequences, and thus can highlight the effect of the actual temporal relationship.

In causal inference, $Y(\bar{T})$ is dependent on both the input video sequence X and the counterfactual temporal relationship $\bar{T}$, i.e., $(X, \bar{T}) \rightarrow Y(\bar{T})$, whereas the original prediction $Y(T)$ is dependent on X and the factual temporal relationship T , i.e., $(X, T) \rightarrow Y(T)$. Thus the effect of the ME temporal relationship can be learned by comparing the difference between the factual prediction Y and its counterfactual intervention $Y(\bar{T})$.

$$Y_{\text{effect}} = Y(T) - Y(\bar{T}). \quad (2)$$

Furthermore, we use the ground-truth as a supervised signal to explicitly guide the counterfactual learning process.

We formulate the joint objective function as

$$\zeta = \zeta_{ce}(Y, y) + \lambda \cdot \zeta_{ce}(Y_{effect}, y), \quad (3)$$

where the symbol y is the ground-truth label, ζ_{ce} is the standard cross-entropy loss, and λ is a hyper-parameter that controls the importance of the counterfactual learning constraint. The joint objective loss optimizes the network in two aspects: 1) the network distinguishes ME by conventional classification loss with the actual correct prediction, i.e., $\zeta_{ce}(Y, y)$; 2) the noteworthy ME temporal relationship is captured through penalizing the results of the wrong ME temporal relationship and highlighting the correct one, i.e., $\zeta_{ce}(Y_{effect}, y)$.

The network model learns the causality between ME temporal variations and ME prediction. Notably, no extra parameters are introduced in the training phase. For MER in the testing phase, the proposed counterfactual learning part is not required and does not introduce extra computational cost.

4 Experiments

In this section, we validate the effectiveness of CoDER on four popular ME datasets. After the implementation details described in Sect. 4.1, we compare CoDER with other state-of-the-art MER methods in Sect. 4.2. Then, we conduct comprehensive ablation studies to analyze the contributions of each component in CoDER in Sect. 4.3. To clearly illustrate the perceived temporal variations in the ME sequence, we present heatmaps of several facial sequences in Sect. 4.4.

4.1 Implementation details

Datasets. We conduct experiments on four publicly MER datasets: SMIC [14], CASME II [15], SAMM [16], and MMEW [5]. SMIC contains 164 ME video clips elicited from 16 subjects. The sequences are annotated in 3 categories, i.e., positive (51 sequences), negative (70 sequences), and surprise (43 sequences). SMIC provides three formats of videos according to recording equipment: high-speed camera (HS) of 100 fps, normal visual camera (VIS) of 25 fps, and near-infrared camera (NIR) of 25 fps. The HS data is used in the experiments, such as those in Refs. [3, 23, 33]. CASME II consists 255 facial sequences from 35 subjects. We use the 5 ME categories in CASME II, i.e., happiness, repression, surprise, disgust, and andothers. Each has 32, 27, 25, 63, and 99 ME videos. SAMM contains 159 facial videos from 27 subjects that are categorized into 8 classes. Following Refs. [3, 13, 24], we exploit the 5 ME classes that contain more than 10 sequences: happiness, anger, surprise, contempt and others. Each has 26, 57, 15, 12, and 26 ME videos. The MMEW contains 300 facial videos elicited from 36 subjects. Following Ref. [5], we use 6 ME categories: happiness, surprise, disgust, fear, angry, and sadness, each of which has 36, 89, 72, 16, 8, and 13 ME videos.

Training. We adopted the preceding three residual blocks in ResNet-50 [44] followed by four transformer layers for ME feature extraction. The input image with a spatial size of 224×224 was encoded to $x \in \mathbb{R}^{1024 \times 14 \times 14}$ via the CNN. We flatten x in the spatial dimension to obtain 196 tokens and map each token to $d = 768$ dimensions with a linear projection layer. Then, a class token is prepended and input to the transformer part. Additionally, learnable position embedding is added to retain positional information. The backbone is pre-trained with the seven basic facial macro-expression images in RAF-DB [45] dataset. We implement all the experiments via PyTorch [46] on a TITAN-RTX GPU with 24 GB of memory. During the training stage, we set the batch size as 16 and the length of a facial sequence as $L = 9$. A facial sequence consists of an apex frame with 8 other frames that are randomly selected center around it, and the placement of 8 frames adheres to their original chronological sequence. The initial learning rate is set as 0.001 and the total training epoch is 80. λ in Eq. (3) is set as 0.5. For R-Counter, we use the first frame to randomly replace the other p frames in an ME sequence. We have used other frames for replacement but obtain no improvement. For D-Counter, we randomly drop a total of p frames in an ME sequence to generate the counterfactually-revised sequences. We set $\lambda = 0.5$ and $p = 3$ via manual grid search. During training, data augmentation, including random cropping, random rotation, color jitter, and random horizontal flip, is applied to each frame. As the database of SMIC is not labeled with apex frames, we use the method proposed in Ref. [17] for spotting the apex frame in each video clip.

Evaluation metric. Following Refs. [4, 5, 13, 17], we use overall the accuracy (ACC) and $F1$ score as the evaluation metrics. The experimental results on the CASME II, SAMM and SMIC datasets are conducted under the leave one subject out (LOSO) protocol. LOSO means that one single subject is left out for evaluation and sequences from other subjects are used as training data. For the MMEW dataset, we adopted the five fold cross-validation evaluation protocol as that in Ref. [5]. For the four datasets, we compare CoDER with the state-of-the-art MER methods that use the same evaluation protocol.

4.2 Comparison with MER methods

Tables 1–4 report the experimental results comparison on MMEW [5], SAMM [16], CASME II [15], and SMIC [14] datasets. Compared with the state-of-the-art MER methods, the proposed CoDER achieves best or second best MER performance. For the MMEW dataset, we follow the experimental results reproduced in Ref. [5].

On the MMEW dataset, CoDER achieves state-of-the-art improvements in overall MER accuracy, as shown in Table 1. Compared with the TLCNN [52] and EL-RCN [21], which exploit LSTM for temporal enrichment,

Table 1 Performance of our proposed counterfactual discriminative micro-expression recognition (CoDER) and state-of-the-art micro-expression recognition (MER) methods on the MMEW [5] dataset. ACC represents accuracy. Bold denotes the best

Method	ACC
ESCSTF [22]	36.6
ResNet10 [44]	36.6
He et al. [47]	54.2
DTSCNN [48]	65.9
ELRCN [21]	41.5
ApexME [49]	48.8
Transfer Learning [50]	52.4
KGSL [51]	56.9
TLCNN [52]	69.4
CoDER	74.8

Table 2 Performance of our proposed CoDER and state-of-the-art MER methods on the SAMM [16] dataset

Method	ACC	F1 score
TSCNN [53]	71.8	0.694
DSSN [23]	57.4	0.464
MSCNN [54]	43.0	0.356
MTMNet [3]	74.1	0.736
LGCcon [17]	40.9	0.340
Graph-TCN [13]	75.0	0.699
GEME [24]	55.9	0.454
MERSiamC3D [26]	68.8	0.640
Lei et al. [55]	74.3	0.705
CoDER	77.9	0.765

Table 3 Performance of our proposed CoDER and state-of-the-art MER methods on the CASME II [15] dataset

Method	ACC	F1 score
DSSN [23]	70.8	0.730
TSCNN [53]	81.0	0.807
MTMNet [3]	75.6	0.701
Graph-TCN [13]	74.0	0.725
TS-AUCNN [33]	72.6	0.670
MSCNN [54]	56.5	0.533
LGCcon [17]	65.0	0.640
Lei et al. [55]	74.3	0.705
GEME [24]	75.2	0.735
CoDER	78.0	0.768

Table 4 Performance of our proposed CoDER and state-of-the-art MER methods on the SMIC [14] dataset

Method	ACC	F1 score
OFF-ApexNet [56]	67.7	0.671
STRCN-G [57]	72.3	0.695
STSTNet [58]	70.1	0.680
DSSN [23]	63.4	0.646
LGCcon [17]	63.4	0.620
TS-AUCNN [33]	76.1	0.710
MTMNet [3]	76.8	0.744
GEME [24]	64.6	0.616
MERSiamC3D [26]	73.6	0.760
CoDER	76.8	0.749

CoDER learns to explicitly perceive subtle facial movements by comparing the difference (normal temporal variation vs. spurious temporal variation) between the normal and counterfactually-revised ME sequences. The improvements of CoDER suggest that explicitly perceiving the temporal relationships with the counterfactual learning mechanism is beneficial as the temporal dynamics in ME usually have a very low intensity. In addition, although numerous MER methods [17, 23, 53, 54] use multi-stream network and take images as well as optical flows for input enrichment, CoDER still outperforms them consistently in terms of overall accuracy and F1 score on the SAMM dataset. Specifically, CoDER outperforms MTMNet, which disentangles ME-related features and transfers knowledge from large amounts of macro-expression training data. CoDER also outperforms Graph-TCN [13] that magnifies ME via an auxiliary magnification network and extracts local facial representations for MER. The superiority of CoDER on the MMEW and SAMM datasets suggests the benefits of the core components in CoDER, i.e., the counterfactual learning module that explicitly perceives the subtle temporal cues in ME. We investigate the contribution of each part in our proposed CoDER in Sect. 4.3.

On the SMIC and CASME II datasets, CoDER is comparable to most MER methods and lags behind TSCNN [53], MERSiamC3D [26] and MTMNet [3]. MERSiamC3D [26] relies on an extra adaptive convolution network to iteratively interpolate the facial frames for each ME clip. As a comparison, CoDER is trained with only the original facial frames as input and does not rely on the cumbersome frame interpolation operation. The improvements of TSCNN [53] over CoDER suggest the benefits of using auxiliary facial action unit (AU) information for MER. We will explore incorporating AU knowledge for MER in future work. Besides, we show the confusion matrix of CoDER on the four ME datasets in Fig. 3. The results illustrate that the ME classes with few facial videos usually have low MER accuracy, e.g., 25 videos for the surprise ME category in CASME II, 25 videos for contempt in SAMM, 8 and 16 videos for anger and fear in MMEW dataset, respectively. We visualize what CoDER “see” when predicting the ME classes in Sect. 4.4.

4.3 Ablation study

Contributions of different components in CoDER. We evaluate the effects of CoDER’s key components in Table 5, including transformer blocks (TB), and two paradigms for generating counterfactually-revised ME sequences: R-Counter, and D-Counter. First, CoDER achieves considerable improvements in MER (ACC: 76.5 vs. 73.7 with R-Counter, 75.5 vs. 73.7 with D-Counter) when incorporating either of the two paradigms for generating the counterfactually-revised ME sequences. MER improvements can be explained in two aspects: 1) CoDER captures

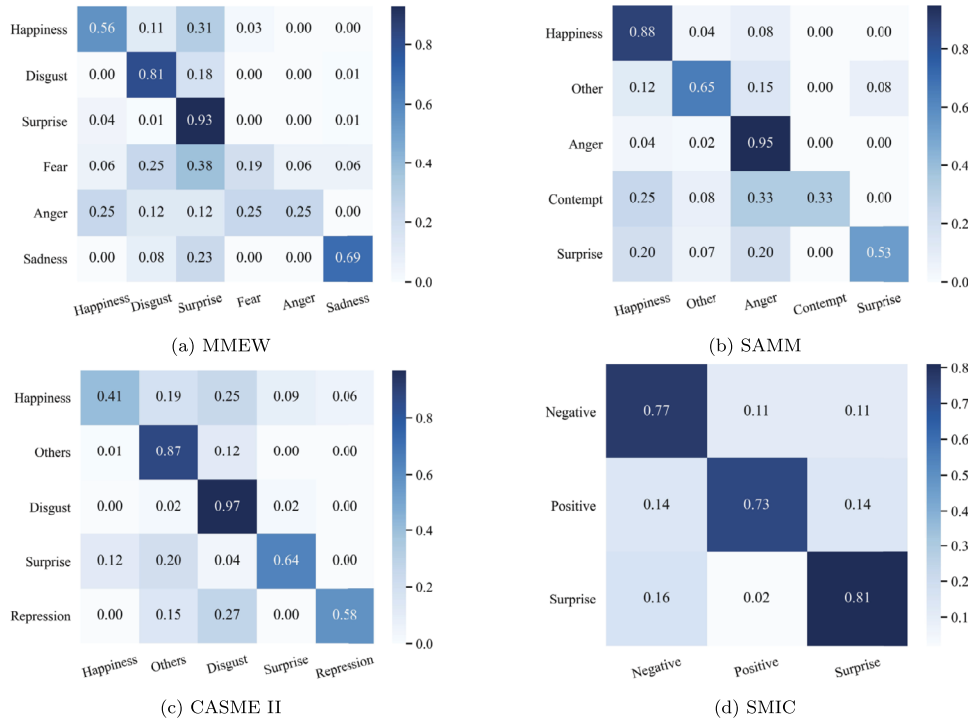


Figure 3 Confusion matrixes of our proposed CoDER on the four ME datasets (MMEW [5], SAMM [16], CASME II [15], and SMIC [14])

the subtle temporal relationships better in ME sequences by training with counterfactually-revised ME sequences; 2) Numerous counterfactually-revised ME sequences have been used for training and facilitate the mitigation of overfitting. Second, CoDER with TB achieves superior MER performance because the self-attention in transformer helps capture the relationships between different facial patches and encode more discriminant spatial features. Besides, the performance improvements achieved by the R-counter are slightly better than that of D-counter. This is because frame dropping ensures only that the discontinuity of the micro-expression intensity changes. In comparison, frame replacement results in a more significantly altered intensity distribution, thereby enabling the neural network to learn a broader range of counterfactual changes.

4.4 Visualization

To clearly illustrate the perceived temporal variations in the ME sequences, we compare the attention maps of CoDER and the baseline model, i.e., CoDER without counterfactual learning modules (CoDER w/o counter).

For illustration convenience, we randomly select four images around the apex frame and the placement of these images adheres to their original chronological sequence. We use the prevalent Grad-CAM [59] method to visualize the input sequential faces. During training for CoDER,

Table 5 Ablation studies on the CASME II dataset. TB:transformer block. R-Counter: frame replacement. D-Counter: frame dropping. ✓ and ✗ denote the use of the method and the removal of the method, respectively

TB	R-Counter	D-Counter	ACC	F1 score
✓	✗	✗	73.7	0.716
✓	✓	✗	76.5	0.736
✓	✗	✓	75.5	0.746
✗	✓	✓	75.9	0.743
✓	✓	✓	78.0	0.768

the ME features derived from the LSTM layer are fed into the counterfactual learning part to encourage the model to perceive the temporal variations. In this manner, the CoDER model is tasked to perceive the subtle temporal characteristics inherent in an ME sequence.

As illustrated in Fig. 4, CoDER shows more meaningful concentration on the input faces and presents obvious temporal variations over time. CoDER shows the greatest concentration on the apex frame and less concentration on the neighbor frames. Moreover, the focused parts of CoDER are consistent with the activated facial action units. This means that CoDER successfully perceives the subtle temporal variation and relationships in the ME sequences, which is critical for accurate MER. In comparison, the baseline method fails to perceive the critical temporal cues as the sequential attention maps show less or

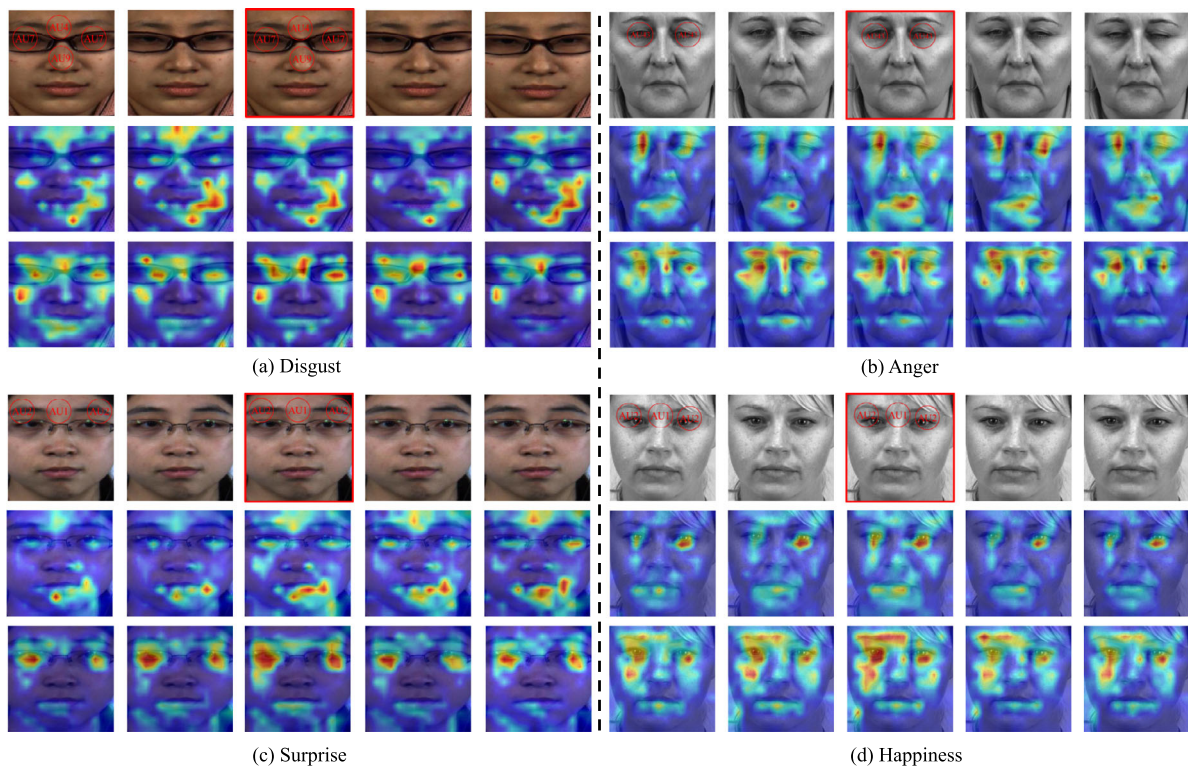


Figure 4 Comparison of attention maps on the CASME II (left) and SAMM (right) datasets. We highlight the apex frame with a black dotted rectangle. Our proposed CoDER perceives the subtle temporal variations of ME better in sub-figures (a)-(d)

no temporal variation. The visualization results in Fig. 4 clearly verify the superiority and feasibility of our proposed CoDER, which can perceive subtle ME temporal cues for robust MER.

5 Conclusion

In this paper, we present an ME counterfactual learning called CoDER framework to encourage the network to perceive subtle temporal cues in the ME video sequence. CoDER is inspired by the idea that ME can be better recognized by observing the temporal facial muscle changes. We propose a counterfactual learning module to perceive subtle temporal cues by comparing the effects of the normal and counterfactually-revised ME sequences. Extensive experiments demonstrate that our proposed CoDER outperforms or is comparable with other state-of-the-art MER methods. The limitations of CoDER are two folds: 1) More counterfactual generation manners can be incorporated, e.g., masking the critical facial regions in the frames to better perceive the local facial deformations; 2) Auxiliary facial action unit knowledge can be incorporated for robust MER. We will explore these points in future work.

Abbreviations

ACC, accuracy; CNN, convolutional neural network; CoDER, counterfactual discriminative micro-expression recognition; LSTM, long short-term memory; ME, micro-expression; MER, micro-expression recognition; TB, transformer blocks.

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Author contributions

All the authors contributed to the study conception and design. The method design and core idea consolidation were performed by YL. The experiments on the four micro-expression datasets were performed by ML, LL and YW. The first draft of the manuscript was written by YL and ZC. All the authors commented on the submitted versions of the manuscript. All the authors read and approved the final manuscript.

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Data availability

The datasets generated during and/or analyzed during the current study are available from the corresponding author upon reasonable request.

Code availability

Our codes are available at https://github.com/sjlrML/Counterfactual_MicroExpression.

Declarations

Competing interests

The authors declare that they have no known financial interests or personal relationships that could influence the work reported in this paper.

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